AIS-PVT: Long-Time AIS Data Assisted Pyramid Vision Transformer for Sea–Land Segmentation in Dual-Polarization SAR Imagery

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***Abstract*— Traditional synthetic aperture radar (SAR) image sea–land segmentation algorithms overlook the ship distribution priori-information provided by the automatic identification system (AIS) data, resulting in poor segmentation performance in complex environments such as ports, marine wetlands, beaches, and other sea–land boundaries. To address the above issues, this article comprehensively uses dual-polarization (VV and VH) SAR images and AIS data as the data source, and it specifically proposes a novel pyramid vision transformer (PVT) assisted by the long-time AIS data (AIS-PVT) for sea–land segmentation. AIS-PVT is the first attempt to integrate the ship distribution density priori-information, provided by the long-time AIS data, into the PVT network, thus the multiscale features of the sea and land can be better distinguished. In the decoding stage, we design a feature filter module (FFM). It aggregates features separately along two spatial directions from the skip connections, enhancing the representation of objects of interest while reducing the influence of redundant information. Furthermore, we develop a boundary-pixel-aware function to steer the model training process, allowing AIS-PVT to concentrate more on the neighborhood information of boundary pixels. Importantly, the AIS-PVT method captures global multiscale information and enhances the model's data fusion capability. The conclusive experimental results demonstrate the superior performance of our approach in sea–land segmentation tasks, outperforming other state-of-the-art (SOTA) techniques.**

***Index Terms*— Automatic identification system (AIS) data assistance, AIS priori-information integrated pyramid vision transformer (AIS-PVT), complex environment, sea–land segmentation, synthetic aperture radar (SAR).**

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I. Introduction

SEA-LAND segmentation is a crucial step in research areas such as ship detection and marine resource management [1], [2]. Accurate sea–land segmentation can eliminate the interference of strong scattering in land regions, effectively reducing the false alarm rate of maritime object detection, and enhancing detection efficiency.

Currently, extensive research has focused on sea–land segmentation in optical remote sensing images [3], [4]. However, these images are susceptible to adverse weather conditions. In contrast, synthetic aperture radar (SAR) imaging offers advantages such as all-weather capability, day-and-night operation, and broad coverage [1]. Therefore, the exploration of sea–land segmentation in SAR images holds substantial practical significance.

Sea–land segmentation methods fall into two main categories: traditional and deep learning-based. Traditional methods, including threshold-based [5], [6], edge detection-based [7], [8], and Markov random fields (MRFs)-based approaches [9], [10], excel in clear sea–land boundaries with less interference. However, they often produce coarse edges and demand substantial manual intervention, making them unsuitable for complex sea–land segmentation tasks.

In recent years, deep learning methods, widely employed in remote sensing tasks like sea ice classification [11], [12], change detection [13], [14], and oil spill detection [15], [16], have been extended to sea–land segmentation. This extension encompasses convolutional neural network (CNN)-based and Transformer-based approaches. Among CNN-based methods, fully convolutional networks (FCNs) have been widely used. For instance, An et al. [17] applied FCN for initial sea–land separation in SAR images. Building upon this, some researchers [18], [19], [20] introduced various attention mechanisms to improve classification accuracy in sea–land segmentation tasks. Lin et al. [21] proposed a multiscale CNN suitable for sea–land segmentation and ship detection, preserving details while enlarging the receptive field. However, FCN networks, with their multiple pooling operations, may lose fine-grained details, limiting their accuracy. In contrast, U-shaped networks use skip connections to fuse high and low-dimensional features, offering notable advantages in sea–land segmentation tasks [22], [23], [24]. For example, Jing et al. [25] introduced the BS-Net, extending U-Net with a boundary

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segmentation interaction module for simultaneous sea–land boundary extraction during segmentation. To enhance the discriminative features for sea and land, while fine-tuning the U-Net network, dual-polarization [26] and multipolarization SAR images [27] were explored for integration during the data input stage. Furthermore, Pelich et al. [28] harnessed distinct backscatter values of different seasonal land features, coupled with Sentinel-1 time series data for coastline detection, thereby enhancing segmentation performance in coastal regions.

Recently, inspired by the success of Transformer in the field of natural language processing (NLP), many researchers have used it in the field of computer vision and called vision transformers (ViTs) [29], [30]. Unlike CNNs, ViTs exhibit certain advantages in handling long-range dependencies and capturing global contextual information. Some researchers have attempted to apply the architecture of ViTs to remote sensing image segmentation and classification [31], [32], [33]. To the best of our knowledge, there are only a few studies that have attempted to apply the architecture of ViTs to sea–land or water–land segmentation tasks. For instance, Yang et al. [34] made preliminary attempts to introduce the Transformer networks into sea–land segmentation tasks, achieving outstanding performance in manual coastline delineation. Zhou et al. [35] devised a U-shaped structure incorporating a structure-aware CNN-Transformer network to address the issue of fine-scale SAR image water body segmentation.

However, deep learning methods also face challenges in sea–land segmentation. Existing models often focus on simpler scenarios, neglecting the complexity of sea–land boundaries in economically active port areas. To address issues like insufficient information in single-polarization data and high noise levels, researchers incorporate different polarization data for improved sea–land discrimination. In SAR images, due to their inherent characteristic, similar backscatter values among different semantic classes pose differentiation challenges. The problem intensifies in complex environments (e.g., ports, marine wetlands, beaches, and other sea–land boundaries) where high backscatter value occlusions induce semantic ambiguity. Relying solely on SAR images proves insufficient in such cases, demanding additional contextual features as semantic cues. Additionally, Current deep-learning methods rarely provide targeted designs for such complex environments.

Based on the above considerations, we have undertaken improvements from both data and network perspectives.

In terms of data enhancement, we make the first attempt to introduce both automatic identification system (AIS) data and dual-polarization SAR data at the network input stage. AIS data offers real-time insights into vessel positions, routes, and navigation details [36]. Ship distribution density information by AIS data, in particular, indicates the concentration of maritime activity, which is invaluable in complex port areas and ship-dense sea regions. In this study, AIS data spanning one year is utilized and referred to as long-time AIS data. Meanwhile, dual-polarization SAR data have proved suitable for sea–land segmentation missions [26], [27], [28]. This combination provides several advantages: it enriches contextual information, aiding in more accurate sea–land discrimination, especially in complex environments. In port regions, artificial structures or vessels may exhibit higher backscattering values, making it challenging to distinguish clear coastlines using SAR data alone. The use of AIS information supplements spatial information for SAR data, which helps to refine segmentation results. The integration of AIS and dual-polarization SAR data also boosts model robustness, mitigating the impact of SAR imaging variations due to weather or climatic conditions.

In terms of network, the above methods are limited in the feature extraction in our multisource and complex environments. Inspired by the effective multiscale feature capturing capabilities of the pyramid ViT (PVT) [37], [38], we propose AIS-PVT, a network-assisted by ship distribution density priori-information from long-time AIS data. AIS-PVT employs a U-shaped architecture, widely used in sea–land segmentation. Furthermore, it innovatively replaces the conventional backbone network with a transformer architecture, marking the first attempt in sea–land segmentation tasks. AIS-PVT leverages the PVT's inherent ability to capture long-range dependencies and contextual information, effectively combining AIS and dual-polarization SAR data. By designing a feature filter module (FFM) after skipping connections, we enhance significant features while reducing interference. Additionally, we develop a boundary-pixel-aware loss function that assigns different weight values to pixels at the sea–land boundary and pixels away from the boundary, ensuring that the model's predictions align with the actual ground truth boundaries.

In summary, our work showcases the potential of combining AIS data and dual-polarization SAR data in the task of sea–land segmentation. The AIS-PVT network maintains superior sea–land segmentation performance in complex environments. The main contributions of this study can be summarized as follows:

1) This is the first attempt to integrate the ship distribution density information, provided by the long-time AIS data, into sea–land segmentation networks. Combined with dual-polarization SAR images, the proposed AIS-PVT can provide richer details and priori information, achieving superior sea–land segmentation, particularly in complex environments like ports, marine wetlands, beaches, and other sea–land boundaries.
2) To enhance feature representation in AIS-PVT, we leverage the pyramid structure of the PVT network, capturing multiscale global context efficiently. Additionally, a special FFM module is designed in the decoder, aggregating features separately along two spatial directions after skip connections. This module is used to refine feature representation, highlighting critical information for sea and land while reducing redundancy.
3) In order to direct the model's attention to sea–land boundary pixels, we develop a boundary-pixel-aware loss function. It assigns adaptive weights to pixels based on differences in importance, enhancing the extraction of precise sea–land boundaries.
4) The proposed AIS-PVT significantly enhances sea–land feature representation by integrating AIS and

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dual-polarization SAR images. It refines multiscale features, enriches feature integrity, and improves boundary delineation precision, as validated by experimental results in complex environments.

The rest of this article is organized as follows. Section II presents our proposed method. Section III introduces the dataset, validates its effectiveness, and performs a comparative analysis with other approaches. Finally, Section IV concludes this article.

II. Methodology

This section first presents an overview of the proposed AIS-PVT architecture, followed by a detailed description of the improved PVT encoder, FFM, and loss function.

A. Overall Architecture

The U-shaped network, known for its versatility in computer vision, combines contextual awareness and precise localization. Proven effective for sea–land segmentation [22], [23], [24], [25], [26], it inspires our simplified AIS-PVT approach, maintaining efficacy without complex architectures.

As shown in Fig. 1, our method employs a four-stage encoder-decoder structure, with the encoder capturing multiscale features based on PVT. We provide three normalized inputs: VV-polarization SAR image, VH-polarization SAR image, and AIS data map. Four pyramid features are then extracted and passed to the decoder stage. Here, four FFMs progressively gather semantic information, preserve details, and filter multiscale features to refine key features. This process enhances the local salient information for the "sea" and "land" categories. After multiple iterations, the final output is a prediction map matching the input size. For training, we propose a boundary-pixel-aware loss function to improve edge fitting between the "sea" and "land" categories. The primary loss optimizes segmentation results by computing the difference between segmentation and ground truth values.

B. PVT Encoder

SAR images often suffer from severe noise due to uncontrollable factors like ground interference and sea surface waves. Additionally, dynamic economic activities in sea regions lead to varying AIS data values. These issues challenge CNNs' feature learning capabilities. Recent research shows that PVT outperforms CNNs in performance and robustness [39], [40]. PVT [37], a pyramid structure, computes representations through spatial reduction attention (SRA) operations for resource efficiency. PVTv2 [Fig. 2(b)] improves on this by replacing the traditional SRA layer with a linear SRA, reducing computational costs. The PVTv2 encoder incorporates three main improvements for dense prediction tasks. First, it integrates overlapping patch embeddings during the patch embedding stage to enhance local continuity features. Second, it adds depthwise separable convolution (DWConv) in the feed-forward stage, uses zero-padding for position encoding, and adapts to feature maps of different resolutions. Third, it replaces the traditional multihead attention (MHA) layer with a linear SRA layer to handle high-resolution images while conserving computational resources, resulting in more powerful feature extraction capabilities.

We adopt PVTv2 as our backbone network to extract robust features for sea–land segmentation, with further modifications for this specific task. We downsize feature maps by a factor of 2 from the input's first layer to capture diverse multiscale semantic information and handle fine details. We balance output feature map channels by starting with 64 channels and doubling them for each subsequent feature map, preventing information imbalance across pyramid levels. As shown in Fig. 1, the original structure includes a classification layer (Softmax function) after each stage (Stages 1–4) for image classification. Since we need to determine the sea or land category for each feature pixel, we remove the classification layer to generate four multiscale feature maps for the input decoding stage. The structure of each output feature map X_i is as follows:

$$X_i \in R^{\frac{H}{2^i}\times\frac{W}{2^i}\times C_i} \tag{1}$$

where i represents the four stages of the encoder, $i \in \{1, 2, 3, 4\}$, and C_i represents the channel number of each output feature map, $C_i \in \{64, 128, 256, 512\}$.

C. Feature Filter Module

Previous algorithms generated numerous redundant features by concatenating low-level and high-level features through skip connections, failing to emphasize regions of interest. To address this, we designed the FFM to process information after skipping connections, integrating features from encoder output and upsampled decoder maps. This module effectively reduces redundancy and enhances the localization of regions of interest.

As shown in Fig. 3, the FFM combines global average pooling (GAP) and global max pooling (GMP). GAP captures global information, aiding the model in perceiving overall features, while GMP captures local importance, enhancing critical details and edges. Integrating these pooling operations helps the model to understand both the overall structure and edge details. Additionally, combining height and width orientation operations with different pooling makes the model more robust to features of varying scales, shapes, and orientations, crucial for accurate segmentation in complex scenes.

For the input $X = [x_1, x_2, x_3, \ldots, x_C] \in \mathbb{R}^{C\times H\times W}$, we employ average pooling kernels $(H_{\text{avg}}, 1)$ and $(1, W_{\text{avg}})$, as well as max pooling kernels $(H_{\max}, 1)$, $(1, W_{\max})$, to separate features in the height and width directions. Therefore, the feature outputs of the cth channel at height h and weight w are as follows, respectively:

$$z_c^h = \frac{1}{W_{\text{avg}}}\sum_{0\le i<W_{\text{avg}}} x_c(h,i) + \frac{1}{W_{\max}}\sum_{0\le i<W_{\max}} x_c(h,i) \tag{2}$$

$$z_c^w = \frac{1}{H_{\text{avg}}}\sum_{0\le j<H_{\text{avg}}} x_c(j,w) + \frac{1}{H_{\max}}\sum_{0\le j<H_{\max}} x_c(j,w). \tag{3}$$

The above obtained the height-direction feature $z_c^h \in \mathbb{R}^{1\times H\times 1}$ and the width-direction feature $z_c^w \in \mathbb{R}^{1\times 1\times W}$ of the

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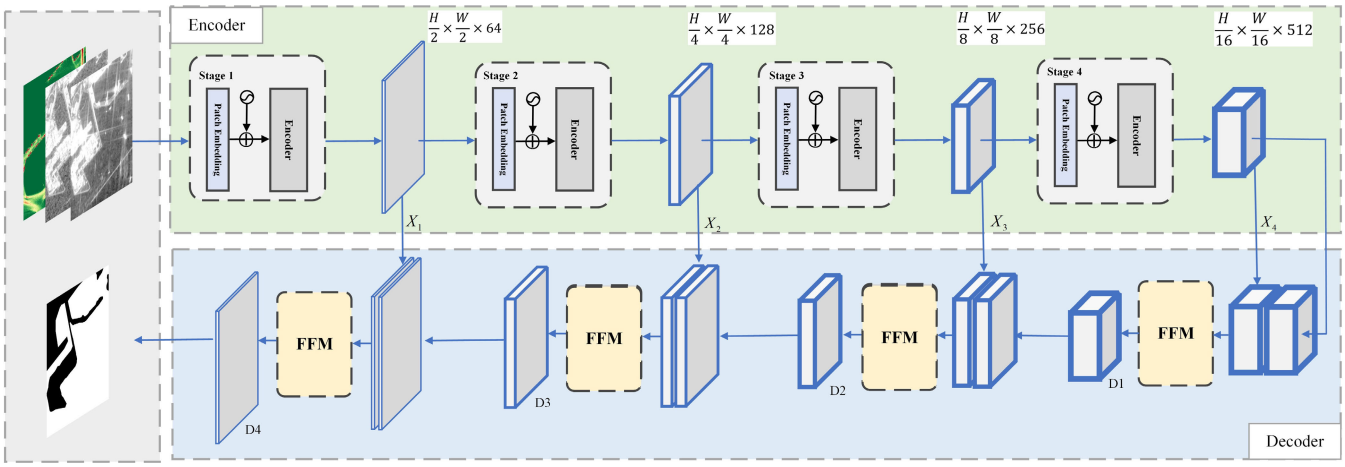


Fig. 1. Architecture of our proposed AIS-PVT.

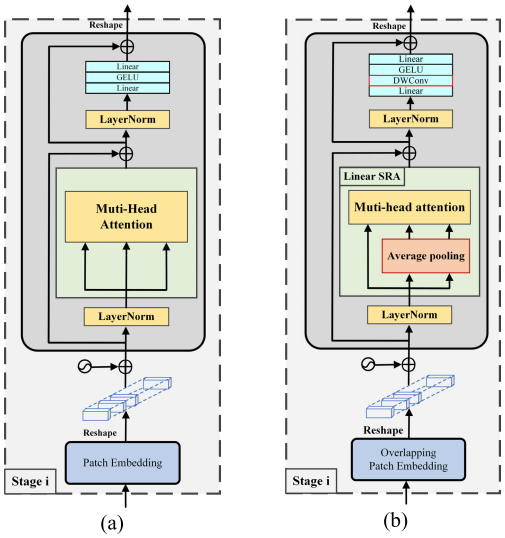


Fig. 2. (a) Structure of the traditional transformer encoder block. (b) Structure of the PVT encoder block.

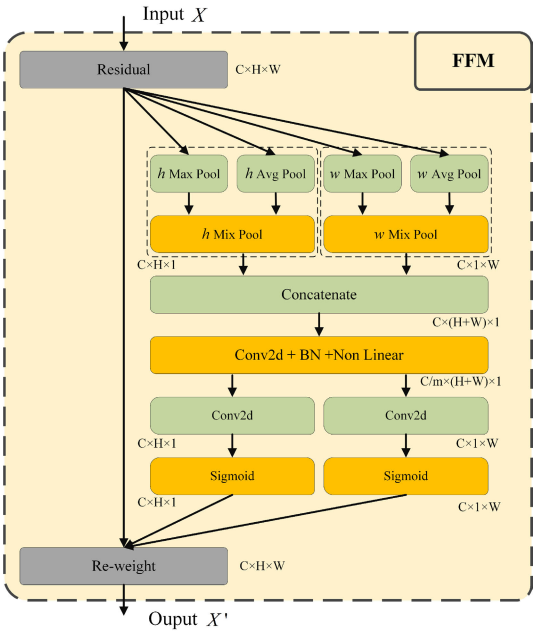


Fig. 3. Illustrative representation of the FFM.

cth channel. After performing the same operation on each channel, we ultimately obtain $Z^h \in \mathbb{R}^{C\times H\times 1}$ and $Z^w \in \mathbb{R}^{C\times 1\times W}$. The above formula combines two pooling operations to gather features from different directions, assisting the model in better understanding the structure and details of the image, and enhancing the model's perception of the spatial structure. This process yields a consolidated feature map, as expressed by the following equation:

$$\boldsymbol{f} = g(F_1(\text{concat}(Z^h, Z^w))) \tag{4}$$

where concat(,) denotes the result of the concatenation operation. F_1 represents a convolutional transformation function with a convolutional kernel of 1×1, it operates exclusively along the channel dimension, integrating information from both directions. $g(\cdot)$ represents the nonlinear activation function, $\boldsymbol{f} \in \mathbb{R}^{C/m\times(H+W)\times 1}$ represents the intermediate feature map, and m denotes the channel reduction ratio.

Subsequently, we reorganize the redundantly filtered feature $\boldsymbol{f}$ into the height and width dimensions, obtaining $\boldsymbol{f}^h \in \mathbb{R}^{C/m\times H\times 1}$ and $\boldsymbol{f}^w \in \mathbb{R}^{C/m\times 1\times W}$. Next, the feature transformation is performed

$$\boldsymbol{f}^{h'} = \sigma(F_h(\boldsymbol{f}^h)) \tag{5}$$

$$\boldsymbol{f}^{w'} = \sigma(F_w(\boldsymbol{f}^w)) \tag{6}$$

where F_h and F_w denote two convolutional transform functions. $\sigma = (1/(1+e^{-x}))$ represent the sigmoid activation function and $\boldsymbol{f}^{h'} \in \mathbb{R}^{C\times H\times 1}$ and $\boldsymbol{f}^{w'} \in \mathbb{R}^{C\times 1\times W}$ represent the final filtering weights, respectively.

Finally, the output X' is enhanced features of the same dimension as X, which can be expressed as follows:

$$X' = X \times \boldsymbol{f}^{h'} \times \boldsymbol{f}^{w'}. \tag{7}$$

The FFM module takes the feature map X as input, combining multiscale features from the encoding phase and upsampling. This introduces many channels with additional features and potential redundancy. Our module focuses attention in both height and width directions, enhancing the model's ability to capture crucial features amidst abundant data. This

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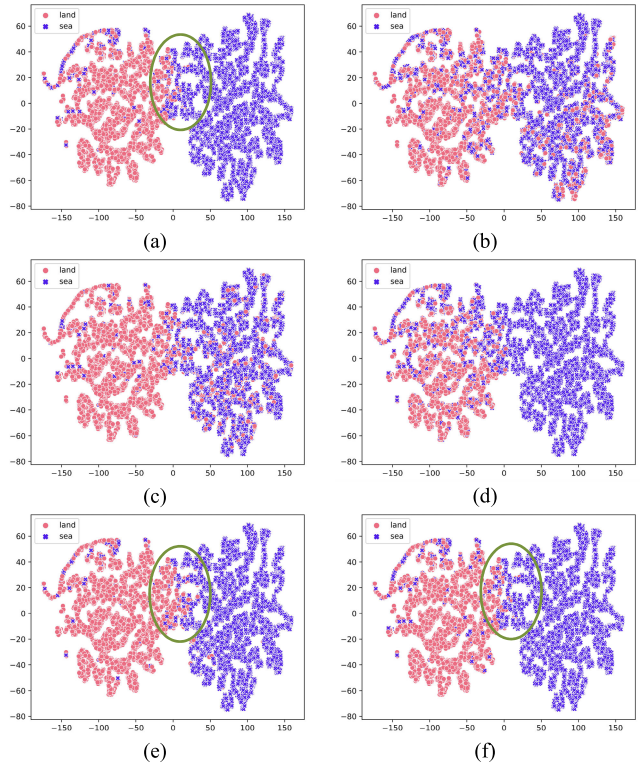


Fig. 4. Influence of FFM module in the decoder on the feature map outputs. (a) Ground truth. (b) AIS-PVT without FFM. (c) AIS-PVT with FFM_1, FFM_1 represents pooling computation conducted solely in the height direction. (d) AIS-PVT with FFM_2, FFM_2 indicates pooling computation performed solely in the width direction. (e) AIS-PVT with FFM_3, FFM_3 signifies the utilization of average pooling computation only during the pooling stage. (f) AIS-PVT with FFM.

improves the model's effectiveness in identifying and classifying sea and land, contributing to overall performance.

To demonstrate the discriminative capability of different components in the FFM module for interclass separability, we use the t-distributed stochastic neighbor embedding (t-SNE) algorithm [41] to visualize the prediction maps, as shown in Fig. 4. FFM_1 represents pooling computation in the height direction, FFM_2 in the width direction, and FFM_3 uses average pooling. Fig. 4 shows that removing any component in the FFM module reduces the differentiation between sea and land. Comparison of data within the green circles [Fig. 4(a), (e), and (f)] reveals that distinguishing regions with similar sea–land features is harder with only average pooling operations. Adding the FFM module enhances the separability between sea and land, effectively filtering distinctive features for both. Fig. 4 demonstrates that the combined effects of each indicator in the proposed module can increase interclass distances. A detailed discussion on the impact of the FFM module on model performance, both quantitatively and qualitatively, is provided in Section III-C3.

D. Boundary-Pixel-Aware Loss Function

In semantic segmentation tasks, the choice of loss function is crucial for training and optimizing network performance. Binary cross-entropy (BCE) loss, commonly used for sea–land segmentation, lacks spatial precision, affecting boundary delineation. Conversely, intersection over union (IoU) loss emphasizes spatial overlap and boundary accuracy but struggles with imbalanced sea–land pixel ratios during training.

To leverage the strengths of both loss functions, we introduce both BCE and IoU and assign varying weights α_{ij} to different pixels. This guides the model's focus toward sea–land boundaries. α_{ij} values are determined based on the difference between the ground truth pixel value and the average value of surrounding pixels, calculated as follows:

$$\alpha_{ij} = e^{\frac{1}{2}\left(\frac{\mathrm{Avg}P_{ij}(G)-G_{ij}}{\lambda}\right)^2} \tag{8}$$

where G_{ij} represents the ground truth value at pixel (i, j) and $\mathrm{Avg}P_{ij}(G)$ represents the average value at pixel (i, j) obtained after applying the average pooling operation on the ground truth.

The absolute difference between $\mathrm{Avg}P_{ij}(G)$ and G_{ij} is calculated. For any given pixel (i, j), a larger difference between $\mathrm{Avg}P_{ij}(G)$ and G_{ij} indicates the pixel point is closer to the sea–land boundary, while a smaller difference suggests that the pixel point is closer to the interior of either the sea or land category. The difference values are then subjected to a Gaussian operation to obtain the final weight α_{ij}. A larger value of α_{ij} assigns a higher weight to pixels near the boundary, contributing more significantly to the loss.

We combine the designed weight values with BCE and IoU to obtain the boundary-pixel-aware function. The calculation formulas are as follows:

$$L_{\alpha\mathrm{BCE}} = \frac{\sum_{i=1}^{H}\sum_{j=1}^{W}\alpha_{ij}[G_{ij}\log(P_{ij})+(1-G_{ij})\log(1-P_{ij})]}{\sum_{i=1}^{H}\sum_{j=1}^{W}\alpha_{ij}} \tag{9}$$

$$L_{\alpha\mathrm{IOU}} = 1 - \frac{\sum_{i=1}^{H}\sum_{j=1}^{W}\alpha_{ij}G_{ij}P_{ij}}{\sum_{i=1}^{H}\sum_{j=1}^{W}\alpha_{ij}(G_{ij}+P_{ij}-G_{ij}P_{ij})} \tag{10}$$

$$\mathrm{Loss} = L_{\alpha\mathrm{BCE}} + L_{\alpha\mathrm{IOU}}. \tag{11}$$

Compared to using only the BCE and IoU functions, our designed loss function strikes a balance between pixel-level accuracy and boundary localization. By assigning varying importance to different pixels, it effectively directs the loss toward the sea–land boundary regions, thereby enhancing the segmentation network's capability to capture intricate boundaries and improving classification accuracy.

III. Experiments and Analysis

A. Datasets for Experiments

We present a sea–land segmentation dataset based on AIS data and dual-polarization SAR images. In the following sections, we offer a detailed introduction to the dataset's sources and the process of sample generation.

1) Source Data and Preprocessing: The Sentinel-1 series satellites, developed and operated by the European Space Agency, are equipped with a C-band SAR system. They offer wide coverage and all-day, all-weather monitoring capabilities [42], enabling high-resolution monitoring of global land, coastal zones, and shipping routes, and large-scale coverage of global oceans. This study used interferometric wide (IW)

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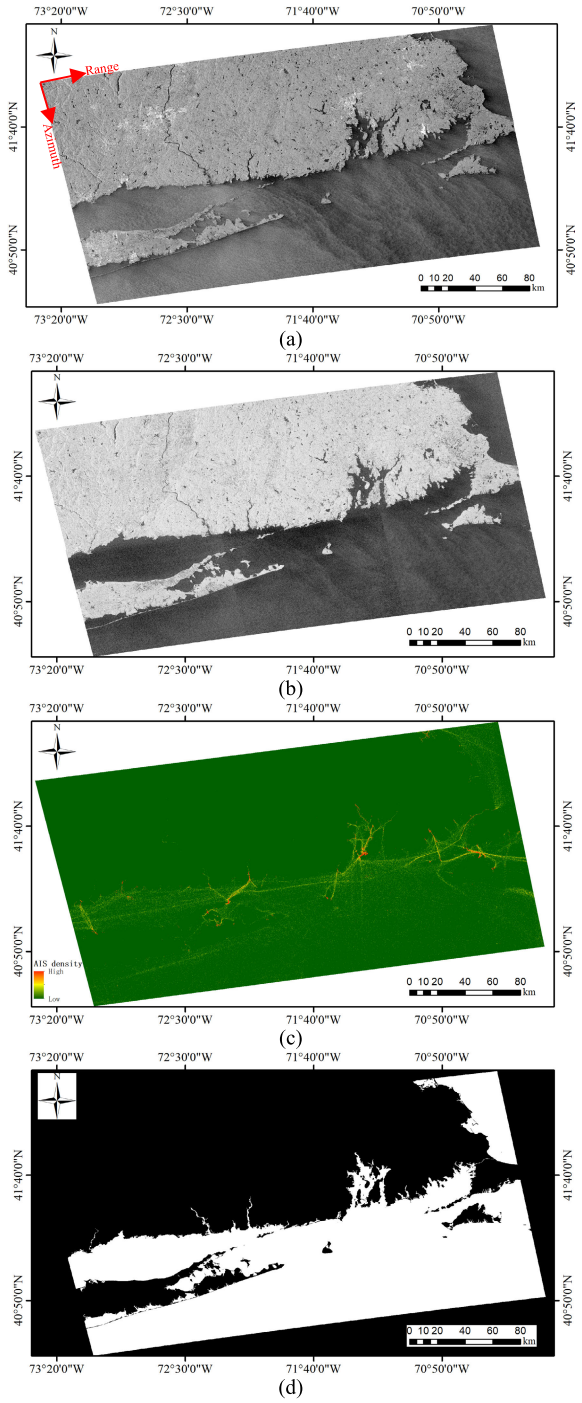


Fig. 5. Example from the dataset. (a) Sentinel-1 VV-polarization image. (b) VH-polarization image. (c) AIS data map. (d) Corresponding segmentation mask showing the pixels divided into "sea" and "land" categories. Each image has a size of 37 882 × 21 171 and a resolution of 10 m.

swath, ground range detected (GRD) images from the Sentinel-1A satellite with dual-polarization (VV, VH). To facilitate the study, the azimuth and range resolutions of the SAR images are standardized to 10 m. Fig. 5(a) and (b) show examples of the preprocessed wide-area Sentinel-1 VV and VH images, respectively.

The AIS is a navigational safety device that transmits real-time positions and characteristics of vessels in sea areas [36]. Raw AIS records are filtered and segmented at one record per minute to describe vessel positions. These records, stored in CSV format and distributed daily, cover all U.S. coastal waters. Ship distribution density increases with the duration of AIS data collection but tends to saturate over time. Therefore, we use AIS data spanning one year, termed long-time AIS data, to balance between short-term information insufficiency and long-term saturation. We conducted frequency statistics on all AIS information within each 10 × 10 m grid unit in the study area over one year, obtaining ship distribution density information at the same resolution as Sentinel-1 data. This processed data, referred to as the AIS data map, is shown in Fig. 5(c), covering the same area as Fig. 5(a).

2) Dataset Labeling and Production Process: The creation of sample labels for the experimental data involves three main steps. First, we apply a thresholding method to the SAR image for the initial binary classification of "sea" and "land." This binary image contains numerous errors and incomplete classifications. Second, we manually refine these results by overlaying AIS data maps and VV- and VH-polarization SAR images, making detailed corrections to sea regions and dense ports using AIS information. Finally, we visually compare the results with optical images from Google Earth and manually correct any remaining misclassified pixels, producing the final segmentation masks for sea and land. Fig. 5 shows examples of large uncropped wide-area images and labels.

The dataset consists of VV-polarization images, VH-polarization images, AIS data maps, and corresponding labels. Due to GPU limitations, we cannot input the entire image into the model for training. To ensure sample representativeness and model stability, we cropped seven wide-area Sentinel-1 images and their corresponding AIS datasets from different geographic regions into 256 × 256 sub-images. Through a series of processing steps, we obtained 4188 pairs for the training set and 900 pairs for the validation set. The ratio of sea to land pixels in the dataset is approximately 1:1. However, it is important to note that this does not imply that every sub-image in the dataset has a 1:1 ratio of land to sea pixels. There are instances of extreme imbalance, such as samples consisting almost entirely of sea regions or almost entirely of land regions. To evaluate the network's performance, we selected a wide-area SAR image not used in the training or validation sets as the test image, which includes various sea and land scenes. This large-scale region spans four state-level administrative areas in the southern US: Louisiana, Mississippi, Alabama, and Florida, and encompasses a diverse array of geographic features such as tidal flats, wetlands, ports, beaches, and rocky coasts. This diversity ensures a rigorous assessment of our method's generalization performance. The dataset is available at https://github.com/DLRS09/SARAIS.

B. Implementation Details and Evaluation Metrics

1) Implementation Details: The proposed model is implemented based on PyTorch. All experiments were conducted on an NVIDIA 3090 GPU with 100 training epochs. Throughout the training process, we employed the AdamW algorithm for network optimization. Additionally, we set the initial learning rate to 0.0001, batch size to 8, and weight decay to 0.0001.

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TABLE I
COMPARISON RESULTS FOR DIFFERENT DATA (%)

Data	Acc	mIoU	iou_0	iou_1	mF1
VV	97.44	94.87	95.76	93.97	97.36
VH	97.49	94.92	95.88	93.95	97.39
VV+VH	98.02	95.99	96.71	95.26	97.95
VV+AIS	98.50	96.02	96.41	95.63	97.95
VH+AIS	98.56	96.50	97.60	95.40	98.20
VV+VH+AIS	**98.69**	**97.33**	**97.82**	**96.84**	**98.64**

TABLE II
DETAILED SETTINGS FOR PVTV2 SERIES

Encoder Backbone	Number of channels of output	Number of heads	Number of layers
PVT v2-B0	[32, 64, 128, 256]	[1, 2, 5, 8]	[2, 2, 2, 2]
PVT v2-B1	[64, 128, 320, 512]	[1, 2, 5, 8]	[2, 2, 2, 2]
PVT v2-B2	[64, 128, 320, 512]	[1, 2, 5, 8]	[3, 4, 6, 3]
PVT v2-ours	**[64, 128, 256, 512]**	**[1, 2, 4, 8]**	**[3, 4, 6, 3]**

2) Evaluation Metrics: We employ five standard accuracy metrics for model evaluation: accuracy, precision, recall, F1-score, and IoU. Accuracy gauges overall correctness, precision assesses positive prediction accuracy, recall evaluates capturing all positives, and F1-score balances precision and recall. IoU, stringent and often yielding lower values, quantifies intersection-to-union proportion. Formulas (12)–(16) detail metric calculations. Additionally, we measure model complexity using the number of parameters, where a larger value indicates a more complex model

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{12}$$

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{13}$$

$$\text{F1} = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \tag{14}$$

$$\text{IoU} = \frac{\text{TP}}{\text{TP} + \text{FP} + \text{FN}} \tag{15}$$

$$\text{ACC} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{FP} + \text{FN} + \text{TN}}. \tag{16}$$

C. Experimental Results and Analysis

1) Segmentation Results in a Large-Scale Scene: We initially applied our method to a large-scale coastal area, using an image with dimensions of 32 768 × 21 504, covering approximately 70 464.31 km^2 [Fig. 6(a)–(c)]. This is the first application of a deep learning model on such a large-scale scene. As shown in Fig. 6(d), our method effectively distinguished sea and land areas, demonstrating clear island boundaries even in busy sea areas and complex ports.

This section provides an overview of our method's performance. In Section III-C2, we explore the impact of different input types on segmentation outcomes. Section III-C3 presents ablation experiments and visualizes the output feature maps. Section III-C4 presents detailed results from representative regions, including regular coastal areas [Region A in Fig. 6(a)], ship-dense sea regions [Region B in Fig. 6(a)], and complex port regions [Region C in Fig. 6(a)], comparing our results with other advanced methods.

2) Validity Analysis of Dataset: In this section, we discuss the impact of different input data on the sea–land segmentation results of our model. To validate the advantages of combining AIS data with dual-polarization (VV and VH) SAR as joint inputs to the model, we explored six different combinations of data, as detailed in Table I.

The quantitative evaluation table and visualized results for the six data combinations are shown in Table I and Fig. 7. Table I reveals that the VV + VH + AIS combination achieved the highest scores across all metrics, followed by VV + AIS and VH + AIS, while the accuracy with only SAR images is lower. This demonstrates that integrating AIS information enhances the model's ability to distinguish between sea and land.

From the SAR images in different regions [Fig. 7(a) and (b)], it is clear that ground interference in busy port areas significantly impacts boundary clarity. Thus, refining segmentation boundaries based solely on SAR images is challenging in complex environments. As shown in Fig. 7(e)–(j), without AIS information, the segmentation boundaries in port areas are incomplete. With AIS data, the model utilizes diverse prior information to accurately determine the sea–land boundary. The VV + AIS and VH + AIS combinations show more false positives and negatives compared to VV + VH + AIS, indicating the importance of both dual-polarization SAR and AIS information in sea–land segmentation.

3) Ablation Studies: We provide a detailed description of the effectiveness of each component in the overall model. To evaluate the effectiveness of the improved encoder, module, and loss, we conducted comparisons by removing or replacing components from the complete AIS-PVT model and comparing them with the standard version.

a) Influences of different encoder backbone: To demonstrate the capability of our improved PVT encoder, we replaced the encoder with three versions of traditional PVTv2, as shown in Table II, with specific differences mainly in the number of heads of the efficient self-attention, number of encoder layers, and number of output channels, respectively. We conducted training comparisons with and without the FFM module for each of the four encoder backbones mentioned above.

As shown in Table III, both PVTv2-B2 and PVTv2-ours achieved the best accuracy results under the presence and absence of the FFM module. However, it is noteworthy that although our accuracy results only slightly outperformed those of B2, our model had fewer parameters compared to B2. This indicates that our encoder achieved higher segmentation performance than B2 with fewer computational resources.

b) Effectiveness of FFM module: We evaluated the FFM module by removing it from the AIS-PVT model and testing different backbones. Table III shows that removing the FFM module decreases ACC, IOU, and F1 values across all models. Fig. 8 visualizes feature outputs from different decoder layers before and after FFM module removal. Without the FFM module, the model captures multiscale global information but lacks detailed sea–land discrimination. With the FFM

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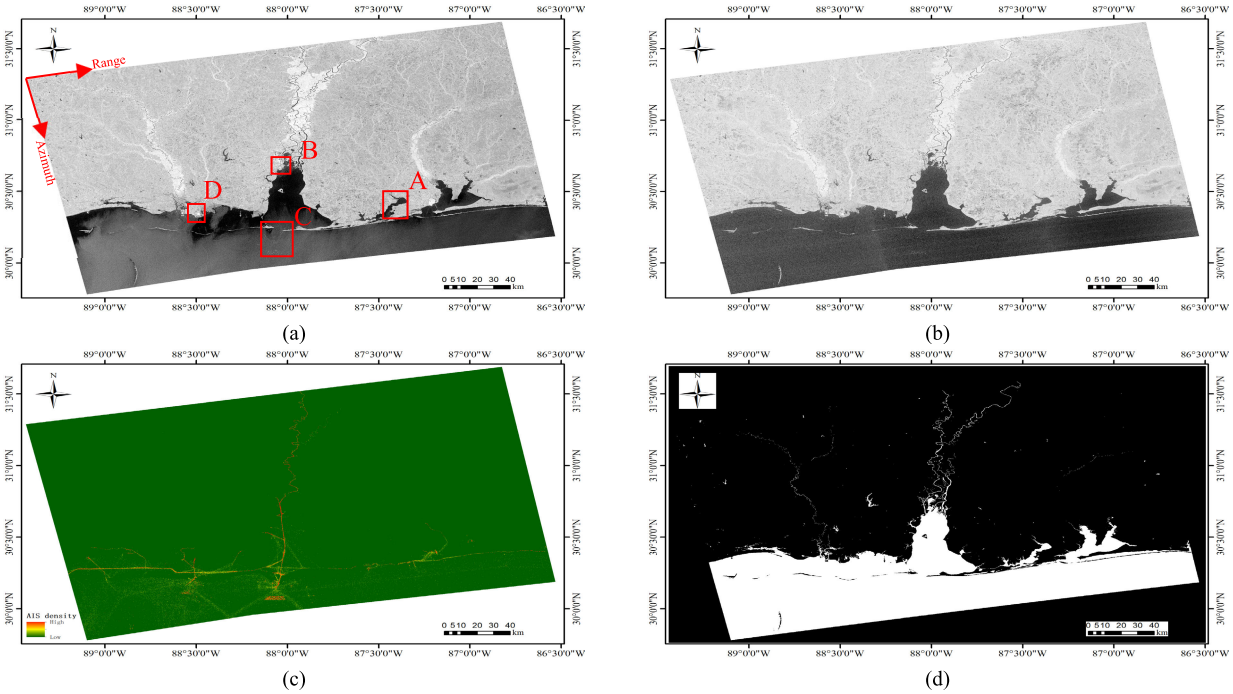


Fig. 6. Visualization results of large-scale scene. (a) VV-polarization image. (b) VH-polarization image. (c) AIS data map. (d) Corresponding segmentation result. This image has a size of 32 768 × 21 504 and a resolution of 10 m. The red boxes A, B, C, and D are four different geographic regions.

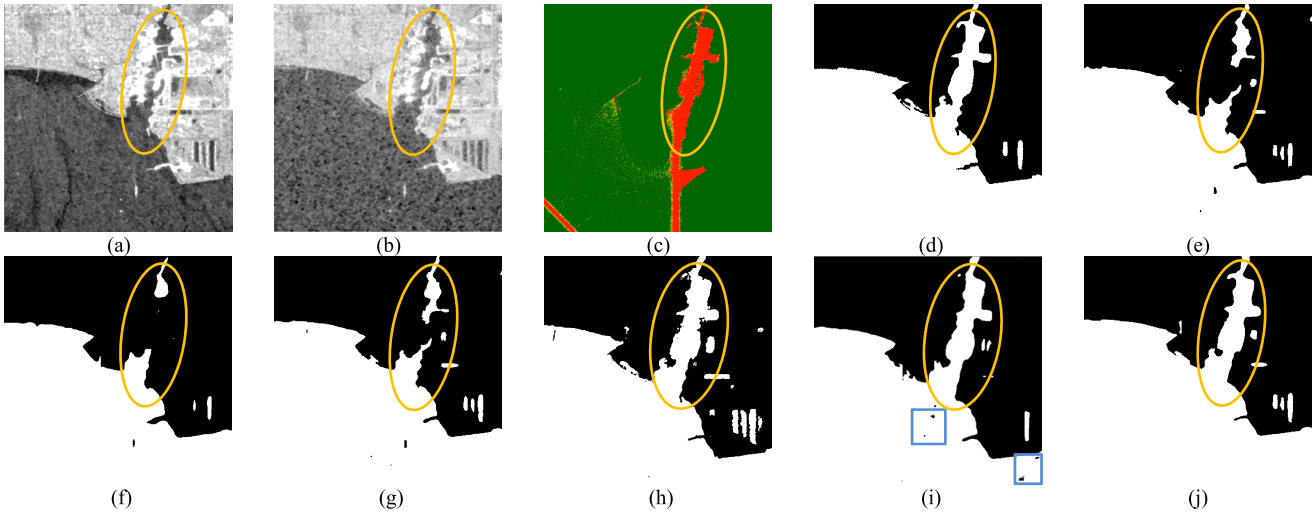


Fig. 7. Visualization results of different input data. (a) Sentinel-1 VV-polarization image. (b) VH-polarization image. (c) AIS data map. (d) Ground truth. (e) VV. (f) VH. (g) VV + VH. (h) VV + AIS. (i) VH + AIS. (j) VV + VH + AIS.

TABLE III

INFLUENCE OF DIFFERENT PVTV2 ENCODERS AND FFM MODULES ON THE RESULTS

Encoder Backbone	Modules	Parameters (M)	Acc (%)	mIoU (%)	iou_0 (%)	iou_1 (%)	mF1 (%)	Recall (%)	Precision (%)
PVT v2-B0	no FFM	**4.93**	97.06	94.57	95.28	94.36	96.99	97.18	96.83
PVT v2-B1		19.58	98.04	95.63	96.24	95.01	97.63	97.69	97.58
PVT v2-B2		30.94	98.06	96.07	96.76	95.38	97.99	98.24	97.78
PVT v2-ours		25.02	98.02	96.48	97.41	95.55	97.95	98.10	97.81
PVT v2-B0	FFM	4.95	98.09	95.62	96.30	94.94	98.02	98.30	97.79
PVT v2-B1		19.63	98.10	96.15	96.86	95.44	98.03	98.14	97.93
PVT v2-B2		30.99	98.22	96.39	97.04	95.75	98.16	98.40	97.95
PVT v2-ours		25.05	**98.69**	**97.33**	**97.82**	**96.84**	**98.64**	**98.82**	**98.49**

module, the model effectively learns distinctive features from SAR images and AIS data, enhancing its ability to identify sea-specific details and edges, underscoring the FFM module's critical role.

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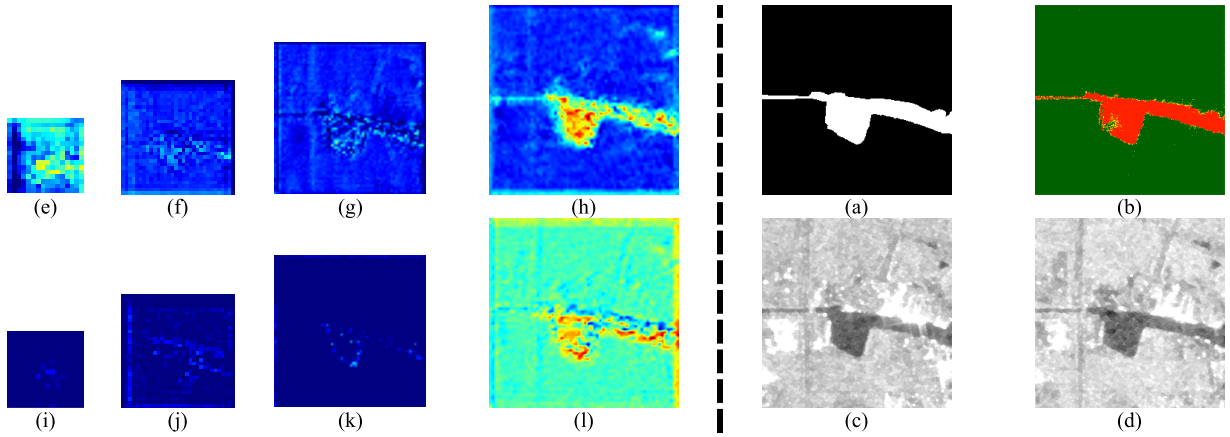


Fig. 8. Influence of FFM module in the decoder on the feature map outputs. (a) Ground truth. (b) AIS data map. (c) Sentinel-1 VV-polarization image. (d) VH-polarization image. (e)–(h) Visual effects of the D1, D2, D3, and D4 feature map outputs using the FFM module are displayed separately. (i)–(l) Visual effects of the D1, D2, D3, and D4 feature map outputs without using the FFM module are displayed separately.

TABLE IV
FITTING PERFORMANCE EVALUATION (%)

Loss Function	Acc	mIoU	iou_0	iou_1	mF1
L_{BCE}	98.01	95.98	96.69	95.26	97.95
L_{IOU}	92.84	86.42	88.12	84.73	92.71
$L_{BCE} + L_{IOU}$	98.04	96.02	96.75	95.29	97.97
$\boldsymbol{L_{\alpha BCE} + L_{\alpha IOU}}$	**98.69**	**97.33**	**97.82**	**96.84**	**98.64**

c) Effectiveness of loss function: Table IV presents the results using various accuracy metrics. Compared to using BCE loss and IoU loss separately, the combined loss function significantly improves segmentation results for the "sea" category but weakens supervision for the "land" category. This is because BCE and IoU treat all pixels equally, ignoring the differences in pixel importance. By assigning a weight α to each pixel, the differences between difficult and simple pixels are amplified, leading to improved IoU for both classes.

4) Detailed Analysis and Comparison With Other Methods: To further demonstrate the effectiveness of our proposed model, we compared it with several state-of-the-art (SOTA) semantic segmentation methods, including U-Net [22], [24], BS-Net [25], HED-UNet [26], and FFEDN [43] for sea–land segmentation, as well as CNN-based methods like DeeplabV3+ [44], PSPNet [45], and U-Net++ [46], and Transformer-based methods such as ST-UNet [31], TransUNet [47], Polyp-PVT [48], and MSGCNet [49]. Table V lists the quantitative accuracy evaluation results of these methods, with the best performance for each metric highlighted in bold. Figs. 9–11 present qualitative segmentation results in regular coastal regions, ship-dense areas, and complex port regions, respectively.

a) Comparison in regular coastal areas: Fig. 9 shows the segmentation results in a typical coastal zone with low AIS data and clear sea–land boundaries in SAR imagery. Most deep learning methods perform well with minimal segmentation noise. However, within the highlighted red boxes, TransUNet, DeepLabV3+, PSPNet, Polyp-PVT, and U-Net++ show varying levels of sea misclassification. FFEDN, HED-UNet, BS-Net, and U-Net improve coastal extraction accuracy but may introduce noise. Notably, ST-UNet, MSGC-Net, and our method, which incorporate attention mechanisms, excel in identifying coastal boundaries and reducing noise, demonstrating the importance of combining Transformer's global context with CNN's local context for precise SAR sea–land segmentation.

b) Comparison in the ship-dense sea region: Fig. 10 shows segmentation results in ship-dense sea regions with active ship activity and rich AIS data. Accurate segmentation here is crucial for subsequent ship recognition and detection. Results indicate that, except for our method and PSPNet, other methods fail to effectively use AIS information, leading to ship misclassification and impacting the reliability of ship recognition. Additionally, FFEDN, HED-UNet, ST-UNet, U-Net++, U-Net, MSGCNet, and BS-Net produce more noise on water surfaces, showing limitations with noisy SAR images. PSPNet captures global context well but lacks shallow feature fusion, resulting in less detailed segmentation and rough sea–land boundaries. In contrast, our method uses PVT for multiscale feature extraction and FFM with spatial and channel attention for accurate segmentation in ship-dense regions.

c) Comparison in the complex port region: Fig. 11 shows the performance of different methods in complex port regions. As noted in Section III-C2, using only SAR imagery for boundary optimization in these regions is impractical due to SAR's inherent limitations. Thus, incorporating AIS data is crucial for accurate sea–land boundary determination. High ship density in these areas correlates strongly with sea–land boundaries, as shown in Figs. 9(c), 10(c), and 11(c). Motivated by this correlation, we integrated AIS information for precise boundary delineation. FFEDN, HED-UNet, BS-Net, and U-Net introduce noticeable noise, while PSPNet struggles with shallow-level information, resulting in inadequate boundary details. ST-UNet and Polyp-PVT show insufficient segmentation completeness. TransUNet combines local and global features, but limited fusion affects detail integration during decoding, causing incomplete results in complex environments. In contrast, our method effectively aggregates discriminative features from both global context

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TABLE V
COMPARISON OF SEGMENTATION RESULTS

Methods	Types	Parameters (M)	Acc (%)	mIoU (%)	iou_0 (%)	iou_1 (%)	mF1 (%)	Recall (%)	Precision (%)
U-Net [22]	CNN	25.31	96.55	93.16	94.25	92.07	96.46	96.99	96.09
BS-Net [25]		25.66	93.15	86.97	88.61	85.33	93.02	94.08	92.66
HED-UNet [26]		8.12	93.25	87.16	88.73	85.60	93.13	94.26	92.79
FFEDN [43]		34.88	93.33	87.30	88.87	85.74	93.21	94.33	92.87
DeepLabV3+ [44]		38.48	97.01	94.01	95.04	92.99	96.91	97.24	96.65
PSPNet [45]		15.24	97.44	94.83	95.79	93.87	97.34	97.39	97.29
U-Net++ [46]		29.36	92.22	85.27	87.37	83.17	92.04	92.72	91.66
ST-UNet [31]	Transformer	160.97	92.70	86.17	87.92	84.41	92.56	93.57	92.20
TransUNet [47]		100.44	98.01	95.96	96.67	95.26	97.94	98.24	97.69
Polyp-PVT [48]		25.11	92.48	85.80	87.50	84.11	92.35	93.50	92.03
MSGCNet [49]		27.02	97.17	91.25	96.57	85.94	95.35	94.47	96.29
Ours	Transformer	25.05	**98.69**	**97.33**	**97.82**	**96.84**	**98.64**	**98.82**	**98.49**

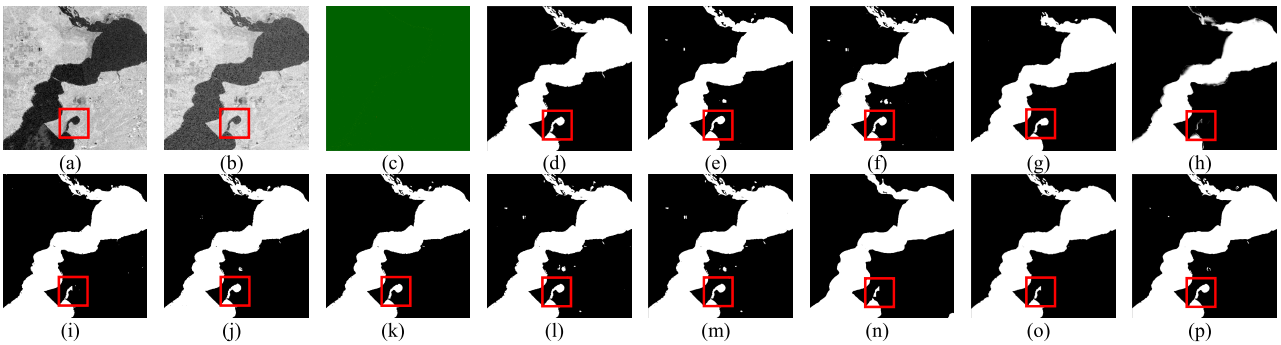


Fig. 9. Visualization results of performance evaluation in regular coastal regions. (a) Sentinel-1 VV-polarization image. (b) VH-polarization image. (c) AIS data map. (d) Ground truth. (e) ST-UNet. (f) TransUNet. (g) MSGCNet. (h) Polyp-PVT. (i) U-Net++. (j) U-Net. (k) BS-Net. (l) FFEDN. (m) HED-UNet. (n) PSPNet. (o) DeepLabV3+. (p) Ours.

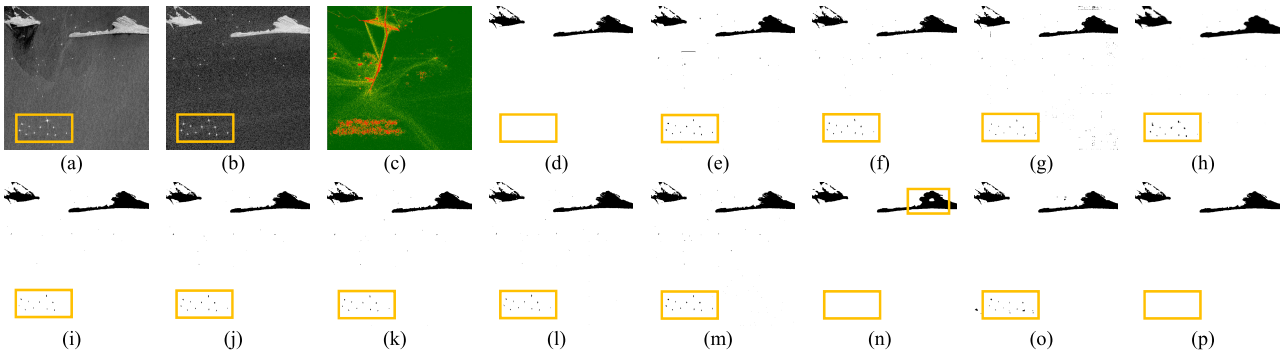


Fig. 10. Visualization results of performance evaluation in ship-dense sea areas. (a) Sentinel-1 VV-polarization image. (b) VH-polarization image. (c) AIS data map. (d) Ground truth. (e) ST-UNet. (f) TransUNet. (g) MSGCNet. (h) Polyp-PVT. (i) U-Net++. (j) U-Net. (k) BS-Net. (l) FFEDN. (m) HED-UNet. (n) PSPNet. (o) DeepLabV3+. (p) Ours.

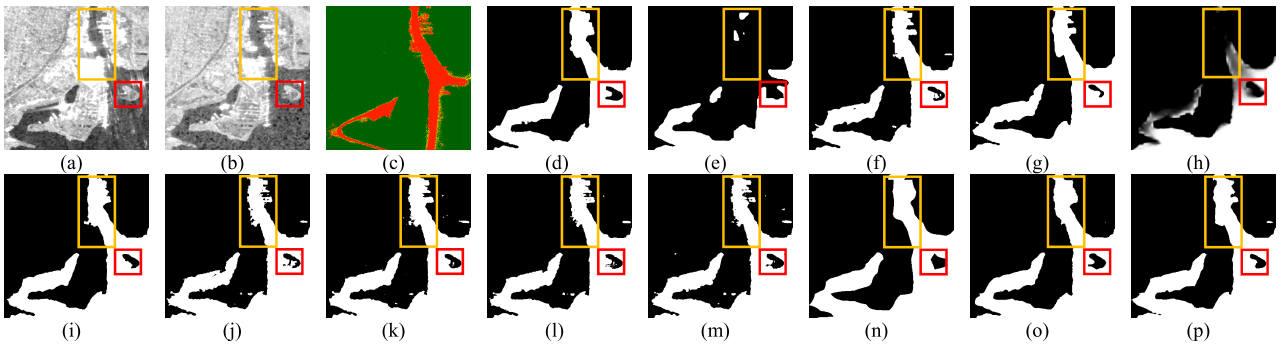


Fig. 11. Visualization results of performance evaluation in complex port regions. (a) Sentinel-1 VV-polarization image. (b) VH-polarization image. (c) AIS data map. (d) Ground truth. (e) ST-UNet. (f) TransUNet. (g) MSGCNet. (h) Polyp-PVT. (i) U-Net++. (j) U-Net. (k) BS-Net. (l) FFEDN. (m) HED-UNet. (n) PSPNet. (o) DeepLabV3+. (p) Ours.

and local features, enabling accurate inferences even in complex scenarios.

Overall, existing methods fail to perform consistently across diverse regions. DeeplabV3+ excels in complex port areas

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with rich AIS data but struggles in areas with moderate AIS data. PSPNet avoids ship misclassification but lacks sea–land boundary sensitivity. TransUNet, Polyp-PVT, MSGCNet, and ST-UNet perform well in low-complexity regions but underperform in ship-dense areas. FFEDN, HED-UNet, BS-Net, and U-Net introduce substantial noise, reducing accuracy. In contrast, our method leverages PVTv2's multiscale encoding to capture global and local context effectively. During decoding, the FFM module fuses features, generating detailed and comprehensive internal features while reducing redundancy, thus improving segmentation accuracy across various regions.

IV. Conclusion

In this study, we addressed the limitations of using only SAR imagery for sea–land segmentation in complex environments, such as ports and ship-dense regions. To overcome this, we integrated ship distribution density from long-term AIS data with dual-polarization SAR images, providing more contextual information and enhancing sea–land discriminative features. Recognizing the challenges existing deep learning methods face with multisource and multiscale data, we proposed AIS-PVT, an improved PVT tailored for sea–land segmentation. AIS-PVT facilitates global multiscale learning and feature extraction, leveraging dual-polarization SAR and AIS data. To integrate channel-wise sensitive features and reduce redundancy, we designed the FFM module, enabling precise segmentation in complex regions. Additionally, we developed a boundary-pixel-aware loss function to address the limitations of existing loss functions in discerning critical pixels at sea–land boundaries, assigning distinct weights to enhance boundary accuracy.

Experimental results show that AIS-PVT significantly improves sea–land feature distinction in dense ship regions and complex ports, enhancing segmentation accuracy in challenging scenarios. Therefore, AIS-PVT holds significant practical value in remote sensing image segmentation. Future work will include adapting AIS information to respond to infrastructure changes and achieving finer sea–land segmentation.

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